

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-I Examination, 2022-23

Botany
Paper: BOB-101 — Cryptogams

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions, selecting at least two each from Sections A and B and Section-C which is compulsory. The figures in the right hand margin indicate marks.

SECTION A

Question 1. Describe the characteristic features of Xanthophyceae. Give an illustrated account of reproduction in *Vaucheria*. [15]

Question 2. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Sexual reproduction in *Ectocarpus*
- (b) Morphology and reproduction of *Scytonema*
- (c) Asexual reproduction in *Volvox*

Question 3. With the help of a labeled diagram, give a brief account of the morphology and sexual reproductive structure of *Marchantia*. [15]

Question 4. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Sexual reproduction in *Anthoceros*
- (b) Cap cell and zoospore formation in *Oedogonium*
- (c) Sporophyte of *Funaria*

SECTION B

Question 5. Give an illustrated account of various types of spore formation in *Puccinia*. [15]

Question 6. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Reproduction in *Rhizopus*
- (b) General characteristics of Basidiomycotina
- (c) Morphology of basidiocarp of *Agaricus*

Question 7. With the help of a suitable diagram, give a brief account of the morphology and reproduction in *Selaginella*. [15]

Question 8. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) T.S. stem of *Equisetum* through internode
- (b) *Cercospora*
- (c) T.S. rachis of *Pteris*

SECTION C

Question 9. Answer the following (not more than 50 words each): [1 × 10 = 10]

- (a) What is the difference between elater and pseudoelater?
- (b) What is the function of vallecular canal?
- (c) Name the fungal group which possesses Crozier.
- (d) What is the function of receptacle in *Sargassum*?
- (e) What are the xerophytic characteristics of *Equisetum* stem?
- (f) Name the Zygosporangium-forming fungi.
- (g) What is dikaryotic phase?
- (h) Name the pigments present in Cyanophyceae members.
- (i) Name the reserved food material present in the Phaeophyceae.
- (j) What is heterokont flagella?

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